

1 Linear algebra

1.1 Vector spaces

We try to avoid presenting a long list of properties, that no one would bother go memorizing tbh. So we start with the concept of group. Recall that a *binary operation* on a set G is just a function $f : G \times G \rightarrow G$. For now on, the familiar operations we have been using since kindergarten (addition, multiplication and scalar multiplication) are simply viewed as functions / binary operations.

Definition 1.1. A **group** is a nonempty set G together with a binary operation on G satisfying the following properties:

1. Associativity property: For all $a, b, c \in G$,

$$a(bc) = (ab)c.$$

2. Two-sided identity: There exists $e \in G$ such that

$$ae = ea = a$$

for all $a \in G$.

3. Inverses: For every $a \in G$, there exists a two-sided inverse element $a^{-1} \in G$ such that

$$a^{-1}a = aa^{-1} = e.$$

Definition 1.2. A group is said to be **abelian** if its binary operation is commutative: $ab = ba$ for all $a, b \in G$.

Definition 1.3. A nonempty set $\mathbb{F}$ equipped with two binary operations on $\mathbb{F}$, addition and multiplication, is a **field** if:

1. $\mathbb{F}$ is an abelian group under addition.
2. $\mathbb{F} - \{0\}$ is an abelian group under multiplication.
3. Multiplication distributes over addition: For all $a, b, c \in \mathbb{F}$, we have $a(b + c)c = ab + ac$.

Definition 1.4. Let $\mathbb{F}$ be a field. A **vector space** over $\mathbb{F}$ is a nonempty set V together with an operation $V \times V \rightarrow V$ on V called **vector addition** (denoted by $+$) and an operation $\mathbb{F} \times V \rightarrow V$ called **scalar multiplication** (denoted by juxtaposition) such that the following properties are satisfied:

1. V is an abelian group under vector addition.
2. Distributivity of scalar multiplication over vector addition: For all $\alpha \in \mathbb{F}$ and $u, v \in V$,

$$\alpha(u + v) = \alpha u + \alpha v.$$

3. Distributivity of scalar multiplication over field addition: For all $\alpha, \beta \in \mathbb{F}$ and $u \in V$,

$$(\alpha + \beta)u = \alpha u + \beta u.$$

4. Compatibility of scalar multiplication with field multiplication: For all $\alpha, \beta \in \mathbb{F}$ and $u \in V$,

$$(\alpha\beta)u = \alpha(\beta u).$$

5. Scalar identity law: $1u = u$ for all $u \in V$.

Definition 1.5. A **subspace** of a vector space V over a field $\mathbb{F}$ is a subset S of V that is a vector space in its own right under the operations obtained by restricting the operations of V to S . Equivalently, S is a nonempty subset that is closed under addition and scalar multiplication, i.e., $x + y \in S$ and $\alpha x \in S$ for all $x, y \in S$ and $\alpha \in \mathbb{F}$. We write $S \leq V$ and say $S < V$ a **proper subspace** if $S \neq V$.

Proposition 1.1. Some facts are presented here (without proof):

1. Examples of field are real or complex numbers. Remark that finite field also exists. if p is a power of a prime, then there exists a field $\mathbb{F}_p$ with p elements. In finite field, we can count the number of elements in a (finite-dimensional) vector space. For example, $\mathbb{F}_p^n$ contains p^n elements.
2. The sum of two subspaces is a subspace. Recall that the sum of two sets is defined by $X + Y = \{x + y \mid x \in X, y \in Y\}$.
3. The intersection of two subspaces is a subspace.
4. The union of two subspaces is **not** necessarily a subspace.
5. The set $\mathbb{F}[x]$ of all polynomials with coefficients in $\mathbb{F}$ equipped with the usual addition of polynomials, and a way of multiplying a polynomial by a scalar $(\alpha p)(x) = \alpha p(x)$ is a vector space. While the set $\mathbb{F}_n[x]$ of all polynomials in $\mathbb{F}[x]$ with degree at most n is a subspace of $\mathbb{F}[x]$.

1.2 Linear independence, basis and dimension

Let V be a vector space over a field $\mathbb{F}$.

Definition 1.6. A nonempty set of vectors S in V is **linearly independent** if for any distinct vectors $s_1, \dots, s_n \in S$,

$$\alpha_1 s_1 + \dots + \alpha_n s_n = 0 \implies \alpha_i = 0 \quad \forall i = 1, \dots, n.$$

Definition 1.7. The **span** of a nonempty subset S , denoted by $\langle S \rangle$, is the set of all linear combinations of elements in S , i.e.,

$$\langle S \rangle = \{\alpha_1 s_1 + \dots + \alpha_n s_n \mid s_i \in S, r_i \in \mathbb{F}\}.$$

If $\langle S \rangle = V$, we say that S **spans** V .

Definition 1.8. A set of vectors S in V is called a **basis** for V if:

1. S is linearly independent.
2. S spans V .

Note that every two bases for V have the same size (even for infinite bases, but we are not going to discuss here). V is **finite-dimensional** if the basis is finite. More precisely, V is n -dimensional and write $\dim V = n$, if the basis has n elements.

Definition 1.9. V is the **direct sum** of two subspaces X and Y , written $V = X \oplus Y$, if

1. $V = X + Y$.
2. $X \cap Y = \{0\}$.

In this case, Y is called a **complement** of X

Proposition 1.2. Some facts are presented here (without proof):

1. (a) $S \subseteq \langle S \rangle$.
 (b) If S is a subset of T , then $\langle S \rangle \subseteq \langle T \rangle$.
 (c) $\langle \langle S \rangle \rangle = \langle S \rangle$.
 (d) The span of two subspaces is equal to their sum, i.e., $\langle X \cup Y \rangle = X + Y$ for all subspaces X, Y .
2. Complements of a subspace are not unique.
3. Let

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ n \end{pmatrix}.$$

Then $\{e_1, \dots, e_n\}$ is called the **standard basis** for $\mathbb{F}^n$.

4. $\{1, x, x^2, \dots, x^n\}$ is a basis for $\mathbb{F}_n[x]$.
5. Every (finite) independent set in a finite-dimensional vector space be extended to a basis. In particular, given subspaces $X < Y < V$. We can find a basis $\{x_1, \dots, x_k, y_1, \dots, y_\ell, v_1, \dots, v_n\}$ for V so that $\{x_1, \dots, x_k\}$ is a basis for X while $\{x_1, \dots, x_k, y_1, \dots, y_\ell\}$ is a basis for Y .
6. If S is a subspace, then $\dim S \leq \dim V$. In particular, $\dim S = \dim V$ implies $S = V$.
7. Let S be a subset of V containing n elements, where $\dim V = n$. If S is linearly independent or $\langle S \rangle = V$, then S is a basis for V .
8. For any two subspaces S and T ,

$$\dim(S + T) + \dim(S \cap T) = \dim S + \dim T.$$

Discussion 1.1 (Counting linearly independent sets in finite fields). We continue the discussion of the finite field in Proposition 1.1.1 (sadly or excitedly, combinatorics again!!!). Instead of counting the number of elements, we wish to determine the number of linear independent sets with exactly k elements ($1 \leq k \leq n$).

- The first element x_1 can be any elements from $\mathbb{F}_p^n$, as long as it is nonzero, so we have $p^n - 1$ choices.
- For the second element x_2 , we want to choose an elements that is not a scalar multiple of x_1 (otherwise they are linearly dependent). Since we have p scalars in $\mathbb{F}_p$, there are p scalar multiples of x_1 . Hence we have $p^n - p$ choices.
- For the i th element x_i , we want $x_1, \dots, x_i$ to be linearly independent. Equivalently, x_i cannot be any linear combination of $x_1, \dots, x_{i-1}$. It is easy to show that there are p^{i-1} linear combinations.

Repeating the argument until we have chosen k elements, we should obtain $(p^n - 1)(p^n - p)(p^n - p^2) \dots (p^n - p^{k-1})$ **ordered** linearly independent sets. Be careful that the sets are ordered (Let's take a simple case, where we find linearly independent sets of size 2 in $\mathbb{F}_2^2$. Following the argument, we have actually counted the sets $\{(1, 0), (0, 1)\}$ and $\{(0, 1), (1, 0)\}$). Hence the number of (unordered) linearly independent sets of with k elements is

$$\frac{(p^n - 1)(p^n - p)(p^n - p^2) \dots (p^n - p^{k-1})}{k!}.$$

1.3 Linear transformations

Definition 1.10. Let V and W be vector spaces over a field $\mathbb{F}$. A function $\psi : V \rightarrow W$ is a **linear transformation** if

$$\psi(\alpha x + \beta y) = \alpha \psi(x) + \beta \psi(y)$$

for all scalars $\alpha, \beta \in \mathbb{F}$ and vectors $x, y \in V$. The set of all linear transformations from V to W is denoted by $\mathcal{L}(V, W)$. We write $\mathcal{L}(V) = \mathcal{L}(V, V)$.

Note that $\psi(\alpha x + \beta y) = \alpha \psi(x) + \beta \psi(y) \forall \alpha, \beta \in \mathbb{F}, x, y \in V$ if and only if $\psi(x + y) = \psi(x) + \psi(y)$ and $\psi(\alpha x) = \alpha \psi(x) \forall \alpha \in \mathbb{F}, x, y \in V$.

Definition 1.11. Let $\psi \in \mathcal{L}(V, W)$. The **kernel** of ψ is

$$\text{Ker } \psi = \{x \in V \mid \psi(x) = 0\}.$$

The **image** of ψ is

$$\text{Im } \psi = \{\psi(x) \mid x \in V\}.$$

Note that $\text{Ker } \psi$ and $\text{Im } \psi$ are subspaces of V and W respectively. So the following definitions make sense:

- $\dim \text{Ker } \psi$ is called the **nullity** of ψ and is denoted by $\text{null } \psi$.
- $\dim \text{Im } \psi$ is called the **rank** of ψ and is denoted by $\text{rank } \psi$.

Proposition 1.3. Some facts are presented (without proof):

1. The sum of two linear transformation is a linear transformation. Similarly, scalar multiples of a linear transformation (here we mean the function is multiplied by elements of $\mathbb{F}$) is a linear transformation. In fact, $\mathcal{L}(V, W)$ is a vector space under two operations stated above (function addition and scalar multiplication).

2. The composition of two linear operators (a name for linear transformation from V to V) in $\mathcal{L}(V)$ is a linear operator.
3. Let $\mathcal{B} = \{b_1, \dots, b_n\}$ be a basis for a vector space V . Given a subset $\{w_1, \dots, w_n\}$ of another vector space W , there exists a unique linear transformation $\psi \in \mathcal{L}(V, W)$ such that $\psi(b_i) = w_i$ for all i . In particular, $\psi_1(b_i) = \psi_2(b_i)$ for all i implies $\psi_1 = \psi_2$. It means that we could easily define a linear transformation by giving its values on the basis.
4. Let $\psi \in \mathcal{L}(V, W)$. Then
 - (a) ψ is surjective if and only if $\text{Im } \psi = W$.
 - (b) ψ is injective if and only if $\text{Ker } \psi = \{0\}$.
 - (c) **Rank-nullity theorem:** If V is finite-dimensional, then

$$\text{rank } \psi + \text{null } \psi = \dim V.$$

Discussion 1.2 (Relation between linear transformations from $\mathbb{F}^n$ into $\mathbb{F}^m$ and matrices). Let $A \in M_{m,n}(\mathbb{F})$. Define a map $\psi_A : \mathbb{F}^n \rightarrow \mathbb{F}^m$ by

$$\psi_A(x) = Ax$$

for all $x \in \mathbb{F}^n$. It can be easily check that ψ_A is a linear transformation from $\mathbb{F}^n$ to $\mathbb{F}^m$. The converse is also true: Any linear transformation from $\mathbb{F}^n$ to $\mathbb{F}^m$ is associated to a matrix. Let $\psi \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ and $\{e_1, \dots, e_n\}$ be the standard basis for $\mathbb{F}^n$. We construct a matrix A as follows:

$$A = (\psi(e_1) \quad \psi(e_2) \quad \cdots \quad \psi(e_n)) \in M_{m,n}(\mathbb{F}).$$

Now we see that for any $x = \alpha_1 e_1 + \cdots + \alpha_n e_n \in \mathbb{F}^n$,

$$\begin{aligned} \psi(x) &= \alpha_1 \psi(e_1) + \cdots + \alpha_n \psi(e_n) \\ &= (\psi(e_1) \quad \psi(e_2) \quad \cdots \quad \psi(e_n)) \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \\ &= A(\alpha_1 e_1 + \cdots + \alpha_n e_n) \\ &= Ax. \end{aligned}$$

Here, A is called the matrix of ψ . Remark that this matrix is unique.

1.4 Ordered bases and change of basis matrices

We have seen there is an order imposed on linearly independent sets so that we can proceed some arguments easily. Now we see some interesting structures when we impose an order on basis.

Definition 1.12. Let V be an n -dimensional vector space over a field $\mathbb{F}$. An **ordered basis** for V is an ordered n -tuple $(b_1, \dots, b_n)$ of vectors for which $\{b_1, \dots, b_n\}$ is a basis for V .

By definition of basis, we know that every $x \in V$ can be expressed as

$$x = \alpha_1 b_1 + \cdots + \alpha_n b_n$$

for some $\alpha_1, \dots, \alpha_n \in \mathbb{F}$. Since $(b_1, \dots, b_n)$ is ordered, x determines a unique ordered n -tuple $(\alpha_1, \dots, \alpha_n)$. Accordingly, let $\mathcal{B} = (b_1, \dots, b_n)$ be an ordered basis, we can define a coordinate map $\phi_{\mathcal{B}} : V \rightarrow \mathbb{F}^n$ by

$$\phi_{\mathcal{B}}(x) = [x]_{\mathcal{B}} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}$$

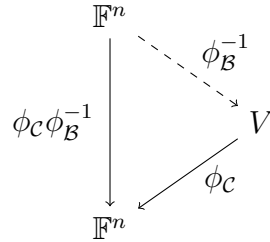
for $x = \alpha_1 b_1 + \cdots + \alpha_n b_n$. We write $[x]_{\mathcal{B}}$ for the **coordinate matrix** of x with respect to $\mathcal{B}$. In fact, it can be shown that $\phi_{\mathcal{B}}$ is a **bijective linear transformation** (vector space isomorphism).

Discussion 1.3 (Change of basis matrix). When we are given an ordered n -tuple $(\alpha_1, \dots, \alpha_n)$ of scalars, the basis uniquely determines a vector $x \in V$ (Those vectors are depending on the basis $\mathcal{B}$). Now a question arises: What happen if we want to represent the vectors using other basis?

Formally speaking, we want to see the relations between coordinate matrices with respect to $\mathcal{B} = (b_1, \dots, b_n)$ and $\mathcal{C} = (c_1, \dots, c_n)$, i.e., we study the map $M : \mathbb{F}^n \rightarrow \mathbb{F}^n$ such that

$$M[x]_{\mathcal{B}} = [x]_{\mathcal{C}}$$

for all $[x]_{\mathcal{B}} \in \mathbb{F}^n$ (of course, for all $x \in V$). Given two coordinate maps $\phi_{\mathcal{B}}$ and $\phi_{\mathcal{C}}$. The following figure



shows that the map that takes $[x]_{\mathcal{B}}$ to $[x]_{\mathcal{C}}$ is of the form $\phi_{\mathcal{C}}\phi_{\mathcal{B}}^{-1}$. Such a map is called the **change of basis operator** and is denoted by $\phi_{\mathcal{B},\mathcal{C}} = \phi_{\mathcal{C}}\phi_{\mathcal{B}}^{-1}$. Note that $\phi_{\mathcal{B},\mathcal{C}}$ is a linear transformation from $\mathbb{F}^n$ to $\mathbb{F}^n$. Hence it follows from Discussion 1.2 that there is a matrix $M_{\mathcal{B},\mathcal{C}}$ associated with $\phi_{\mathcal{B},\mathcal{C}}$ and we call it the **change of basis matrix** from $\mathcal{B}$ to $\mathcal{C}$. We can determine the structure of $M_{\mathcal{B},\mathcal{C}}$ in terms of $\mathcal{B}$ and $\mathcal{C}$. To see this,

$$\begin{aligned} M_{\mathcal{B},\mathcal{C}} &= (\phi_{\mathcal{B},\mathcal{C}}(e_1) \quad \cdots \quad \phi_{\mathcal{B},\mathcal{C}}(e_n)) \\ &= (\phi_{\mathcal{C}}\phi_{\mathcal{B}}^{-1}(e_1) \quad \cdots \quad \phi_{\mathcal{C}}\phi_{\mathcal{B}}^{-1}(e_n)) \\ &= (\phi_{\mathcal{C}}\phi_{\mathcal{B}}^{-1}([b_1]_{\mathcal{B}}) \quad \cdots \quad \phi_{\mathcal{C}}\phi_{\mathcal{B}}^{-1}([b_n]_{\mathcal{B}})) \quad \because [b_i]_{\mathcal{B}} = e_i \quad \forall i \\ &= (\phi_{\mathcal{C}}(b_1) \quad \cdots \quad \phi_{\mathcal{C}}(b_n)) \\ &= ([b_1]_{\mathcal{C}} \quad \cdots \quad [b_n]_{\mathcal{C}}). \end{aligned}$$

Note also that both coordinate maps are isomorphisms and thus the change of basis operator $\phi_{\mathcal{C},\mathcal{B}}$ from $\mathcal{C}$ to $\mathcal{B}$ can be determined as follows:

$$\phi_{\mathcal{C},\mathcal{B}} = \phi_{\mathcal{B}}\phi_{\mathcal{C}}^{-1} = (\phi_{\mathcal{C}}\phi_{\mathcal{B}}^{-1})^{-1} = \phi_{\mathcal{B},\mathcal{C}}^{-1}.$$

So the composition $\phi_{\mathcal{C},\mathcal{B}}\phi_{\mathcal{B},\mathcal{C}}$ is the identity map, which implies that $M_{\mathcal{C},\mathcal{B}}M_{\mathcal{B},\mathcal{C}} = I_n$ and thus $M_{\mathcal{C},\mathcal{B}} = M_{\mathcal{B},\mathcal{C}}^{-1}$.

Discussion 1.4 (Matrix of a linear transformation). We have known the matrix of a linear transformation from $\mathbb{F}^n$ to $\mathbb{F}^m$ in Discussion 1.2. Now, we want to generalize the result to study the matrix of a linear transformation between finite-dimensional vector spaces. Let $\mathcal{B} = (b_1, \dots, b_n)$ and $\mathcal{C}$ be ordered bases for vector spaces V and W respectively, with $\dim V = n$ and $\dim W = m$. Let $\psi \in \mathcal{L}(V, W)$. Then the map $\theta : \mathbb{F}^n \rightarrow \mathbb{F}^m$ defined by

$$\theta([x]_{\mathcal{B}}) = [\psi(x)]_{\mathcal{C}}$$

is a linear transformation from $\mathbb{F}^n$ to $\mathbb{F}^m$ (we can view it as a representation of ψ). Hence there is a matrix A such that

$$A[x]_{\mathcal{B}} = [\psi(x)]_{\mathcal{C}}.$$

Since $[b_i]_{\mathcal{B}} = e_i$ for all i , we have $A^{(i)} = Ae_i = [\psi(b_i)]_{\mathcal{C}}$, where $A^{(i)}$ denotes the i th column of A . So A is of the form

$$A = ([\psi(b_1)]_{\mathcal{C}} \quad \cdots \quad [\psi(b_n)]_{\mathcal{C}}).$$

The matrix A , now is called the matrix of ψ with respect to $\mathcal{B}$ and $\mathcal{C}$, and is denoted $[\psi]_{\mathcal{B},\mathcal{C}}$. If $W = V$ and $\mathcal{C} = \mathcal{B}$, then we write $[\psi]_{\mathcal{B}} = [\psi]_{\mathcal{B},\mathcal{B}}$.

When $\mathcal{B}$ and $\mathcal{C}$ are fixed, we are wondering whether there is a one-to-one correspondence between linear transformations and matrices. It turns out that the answer is positive. We define a map $\mu : \mathcal{L}(V, W) \rightarrow M_{m,n}(\mathbb{F})$ by

$$\mu(\psi) = [\psi]_{\mathcal{B},\mathcal{C}}.$$

We can verify that μ is an isomorphism and hence $\mathcal{L}(V, W) \approx M_{m,n}(\mathbb{F})$. Moreover, let $\mathcal{D}$ be an ordered basis for a k -dimensional vector space U and $\varphi \in \mathcal{L}(W, U)$. The matrix of the composition $\varphi \circ \psi$ is just the product of the matrices of φ and ψ . To prove this, let $x \in V$. Then we have

$$\begin{aligned} [\varphi \circ \psi]_{\mathcal{B},\mathcal{D}}[x]_{\mathcal{B}} &= [(\varphi \circ \psi)(x)]_{\mathcal{D}} \\ &= [(\varphi(\psi(x)))_{\mathcal{D}} \\ &= [\varphi]_{\mathcal{C},\mathcal{D}}[\psi(x)]_{\mathcal{C}} \\ &= [\varphi]_{\mathcal{C},\mathcal{D}}[\psi]_{\mathcal{B},\mathcal{C}}[x]_{\mathcal{B}}. \end{aligned}$$

So $[\varphi \circ \psi]_{\mathcal{B},\mathcal{D}} = [\varphi]_{\mathcal{C},\mathcal{D}}[\psi]_{\mathcal{B},\mathcal{C}}$ as desired.

When $[\psi]_{\mathcal{B},\mathcal{C}}$ is found, describing ψ with respect to another pair of ordered bases $\mathcal{B}'$ and $\mathcal{C}'$ would be easier, i.e., by using

$$[\psi]_{\mathcal{B}',\mathcal{C}'} = M_{\mathcal{C},\mathcal{C}'}[\psi]_{\mathcal{B},\mathcal{C}}M_{\mathcal{B},\mathcal{B}'}^{-1}.$$

The following are figures to summarize the discussion.

$$\begin{array}{ccc}
V & \xrightarrow{\psi} & \psi(V) \xrightarrow{\varphi} \varphi(\psi(V)) \\
\downarrow \phi_{\mathcal{B}} & & \downarrow \phi_{\mathcal{C}} \quad \downarrow \phi_{\mathcal{D}} \\
\mathbb{F}^n & \xrightarrow{[\psi]_{\mathcal{B},\mathcal{C}}} & \mathbb{F}^m \xrightarrow{[\varphi]_{\mathcal{C},\mathcal{D}}} \mathbb{F}^k
\end{array}
\quad
\begin{array}{ccc}
[x]_{\mathcal{B}'} & \xrightarrow{M_{\mathcal{B}',\mathcal{B}}} & [x]_{\mathcal{B}} \\
\downarrow [\psi]_{\mathcal{B}',\mathcal{C}'} & & \downarrow [\psi]_{\mathcal{B},\mathcal{C}} \\
[\psi(x)]_{\mathcal{C}'} & \xleftarrow{M_{\mathcal{C},\mathcal{C}'}} & [\psi(x)]_{\mathcal{C}}
\end{array}$$

$$[\varphi \circ \psi]_{\mathcal{B},\mathcal{D}} = [\varphi]_{\mathcal{C},\mathcal{D}} [\psi]_{\mathcal{B},\mathcal{C}}$$

Discussion 1.5 (Rank of a linear operator). In this discussion, we just solve a rank problem using many approaches: Let τ be a linear operator on a finite-dimensional vector space V . Suppose that there is a positive integer n such that $\text{rank } \tau^n = \text{rank } \tau^{n+1}$. Show that $\text{rank } \tau^n = \text{rank } \tau^{n+k}$ for all $k \geq 0$.

Approach 1. Transform it into a kernel problem: By rank-nullity theorem, the hypothesis is equivalent to $\text{null } \tau^n = \text{null } \tau^{n+1}$, which means that $\text{Ker } \tau^n = \text{Ker } \tau^{n+1}$. Cases $k = 0, 1$ are done. Suppose $k \geq 2$. We want to show $\text{Ker } \tau^{n+k} \subseteq \text{Ker } \tau^n$ since it is clear that $\text{Ker } \tau^n \subseteq \text{Ker } \tau^{n+k}$.

Let $x \in \text{Ker } \tau^{n+k}$. We can apply induction if we can show that $x \in \text{Ker } \tau^{n+k-1}$. To achieve the aim,

$$\tau^{n+k}x = 0 \implies \tau^{n+1}\tau^{k-1}x = 0 \implies \tau^{k-1}x \in \text{Ker } \tau^{n+1}.$$

Since $\text{Ker } \tau^{n+1} = \text{Ker } \tau^n$, we have

$$\tau^{k-1}x \in \text{Ker } \tau^n \implies \tau^n \tau^{k-1}x = 0 \implies x \in \text{Ker } \tau^{n+k-1}.$$

Then we are done! □

Approach 2. Transform it into a matrix problem: In view of Discussion 1.4, we can identify τ as a square matrix A . So the problem statement becomes $\text{rank } A^n = \text{rank } A^{n+1} \implies \text{rank } A^n = \text{rank } A^{n+k}$ for all $k \geq 0$. We still apply induction here. The base case is done. Assume that $k \geq 2$ and $\text{rank } A^n = \text{rank } A^{n+\ell}$ holds for all $0 \leq \ell < k$. Clearly, $\text{rank } A^{n+k} \leq \text{rank } A^n$. For the converse, we use the **Frobenius inequality**:

$$\text{rank } XY + \text{rank } YZ \leq \text{rank } Y + \text{rank } XYZ.$$

In our case, setting $X = A^{k-1}$, $Y = A^n$ and $Z = A$ yields

$$\text{rank } A^{n+k-1} + \text{rank } A^{n+1} \leq \text{rank } A^n + \text{rank } A^{n+k}.$$

By induction hypothesis, we have $\text{rank } A^{n+k-1} = \text{rank } A^n$ and $\text{rank } A^{n+1} = \text{rank } A^n$. So we get $\text{rank } A^n \leq \text{rank } A^{n+k}$, completing the proof. □

Remark. I try to explain the motivation behind choosing such X, Y, Z : First we know our aim is $\text{rank } A^n \leq \text{rank } A^{n+k}$. The relation $\leq$ suggests that we might choose $Y = A^{n+k}$ or $XYZ = A^{n+k}$ (as they are on right hand side). Note that A is not necessarily invertible, so we decide to take $XYZ = A^{n+k}$. Next, we decompose $A^{n+k} = A^a A^b A^c$, where $a, b, c \geq 1$. We see $a, b, c < n+k$, and hence they can apply induction hypothesis after we settle all small issues. For this purpose, we pick $X = A^a$, $Y = A^b$ and $Z = A^c$ with $a+b, b+c, a+c \geq n$. We can just choose one number to simplify the calculations, like setting $c = 1$ as in our solution.

Any other approaches are welcome!

1.5 Determinants and Laplace expansion theorem

The following definition is not important.

Definition 1.13. Let $A = (a_{ij}) \in M_n(\mathbb{F})$, where $\mathbb{F}$ is a field. The **determinant** of A , denoted by $\det A$, is defined as

$$\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)} \cdots a_{n\sigma(n)},$$

where S_n is the symmetric group on $\{1, \dots, n\}$. In other words, the sum ranges over all the permutations of $\{1, \dots, n\}$. We also use the notation $\pi_\sigma(A) = a_{1\sigma(1)} \cdots a_{n\sigma(n)}$ so that

$$\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \pi_\sigma(A).$$

It seems to be complicated, but we have several studies that improved the calculation of determinants, i.e., Laplace expansion theorem.

Definition 1.14. Let $A \in M_{m,n}(\mathbb{F})$, $1 \leq i_1 < \cdots < i_s \leq m$, $1 \leq j_1 < \cdots < j_t \leq n$. For index sets $\alpha = \{i_1, \dots, i_s\} \subseteq \{1, \dots, m\}$ and $\beta = \{j_1, \dots, j_t\} \subseteq \{1, \dots, n\}$, denote by $A[\alpha, \beta] = A[i_1, \dots, i_s | j_1, \dots, j_t]$ the $s \times t$ **submatrix** of A that lies in the rows indexed by α and the columns indexed by β . If $\alpha = \beta$, the submatrix $A[\alpha, \alpha]$ is called a principal submatrix of A . Also, α^c denotes $\{1, \dots, m\} \setminus \alpha$.

Theorem 1.1 (Expansion along a row or a column). Let $A_{st} = A[\{s\}^c, \{t\}^c]$ for indices s, t . Then for fixed indices i, j , we have

$$\det A = \sum_{k=1}^n (-1)^{i+k} a_{ik} \det A_{ik}$$

and

$$\det A = \sum_{k=1}^n (-1)^{k+j} a_{kj} \det A_{kj}.$$

The first sum is the **Laplace expansion along row i** , while the second sum is the **Laplace expansion along column j** .

Finish? Not yet huhu. The theorem above is just a special case of General Laplace expansion theorem.

Theorem 1.2 (General Laplace expansion theorem). Let $1 \leq k \leq n$ be an integer, and let J be an index set of $\{1, \dots, n\}$ with $|J| = k$. Then

$$\begin{aligned} \det A &= \sum_{I \subseteq \{1, \dots, n\}, |I|=k} (-1)^{p(I, J)} \det A[J, I] \det[J^c, I^c] \\ &= \sum_{I \subseteq \{1, \dots, n\}, |I|=k} (-1)^{p(I, J)} \det A[I, J] \det[I^c, J^c] \end{aligned}$$

where $p(I, J) = \sum_{i \in I} i + \sum_{j \in J} j$.

Remark that the sums are over all index sets I with $|I| = k$. Choosing $J = \{i\}$ or $\{j\}$ gives the expansion in Theorem 1.1.

Discussion 1.6 (Vandermonde matrix). We only discuss square Vandermonde matrices because they have determinant. A Vandermonde matrix is an $n \times n$ matrix

$$V = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ x_1^2 & x_2^2 & \cdots & x_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_1^n & x_2^n & \cdots & x_n^n \end{pmatrix}.$$

The determinant (Vandermonde determinant) is given by

$$\det V = \sum_{1 \leq i < j \leq n} (x_j - x_i).$$

Sometimes we need some luck to spot this matrix in a problem, as it won't always be immediately become so obvious. Let's try a problem using Vandermonde matrix. Given an integer $n \geq 3$. Suppose that $g(x) \in \mathbb{C}[x]$ with $\deg g(x) = n$. If $g(x)$ has n distinct roots $z_1, \dots, z_n$, show that

$$\sum_{i=1}^n \frac{f(z_i)}{g'(z_i)} = 0$$

for every polynomial $f(x) \in \mathbb{C}[x]$ with $\deg f(x) \leq n-2$. Here, $g'(x)$ denotes the derivative of $g(x)$.

Proof. With the given informations, we could write $g(x) = c(x - z_1) \cdots (x - z_n)$ for some nonzero $c \in \mathbb{C}$. Applying the chain rule gives

$$g(x) = c \sum_{k=1}^n \prod_{i \neq k} (x - z_i).$$

So for each root z_ℓ , we see that

$$\begin{aligned} g'(z_\ell) &= c \sum_{k=1}^n \prod_{i \neq k} (z_\ell - z_i) \\ &= c \prod_{i \neq \ell} (z_\ell - z_i) \quad \because \prod_{i \neq k} (z_\ell - z_i) = 0 \quad \forall k \in \{1, \dots, n\} \setminus \{\ell\}. \end{aligned}$$

Now it can be expressed using Vandermonde determinant as follows:

$$\begin{aligned} g'(z_\ell) &= c(z_\ell - z_1) \cdots (z_\ell - z_{\ell-1}) \underbrace{(z_\ell - z_{\ell+1}) \cdots (z_\ell - z_n)}_{\text{Take out } -1} \\ &= (-1)^{n-\ell} c(z_\ell - z_1) \cdots (z_\ell - z_{\ell-1})(z_{\ell+1} - z_\ell) \cdots (z_n - z_\ell) \\ &= \frac{(-1)^{n-\ell} c(z_\ell - z_1) \cdots (z_\ell - z_{\ell-1})(z_{\ell+1} - z_\ell) \cdots (z_n - z_\ell) \prod_{1 \leq i < j \leq n, i, j \neq \ell} (z_j - z_i)}{\prod_{1 \leq i < j \leq n, i, j \neq \ell} (z_j - z_i)} \\ &= c(-1)^{n-\ell} \frac{\prod_{1 \leq i < j \leq n} (z_j - z_i)}{\prod_{1 \leq i < j \leq n, i, j \neq \ell} (z_j - z_i)} \\ &= c(-1)^{n-\ell} \frac{\det V}{\det V_\ell} \end{aligned}$$

where

$$V = \begin{pmatrix} 1 & z_1 & z_1^2 & \cdots & z_1^{n-1} \\ 1 & z_2 & z_2^2 & \cdots & z_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & z_n & z_n^2 & \cdots & z_n^{n-1} \end{pmatrix}, \quad V_\ell = \begin{pmatrix} 1 & z_1 & z_1^2 & \cdots & x_1^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & z_{\ell-1} & z_{\ell-1}^2 & \cdots & x_{\ell-1}^{n-2} \\ 1 & z_{\ell+1} & z_{\ell+1}^2 & \cdots & z_{\ell+1}^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & z_n & z_n^2 & \cdots & z_n^{n-2} \end{pmatrix}.$$

Therefore,

$$\sum_{\ell=1}^n \frac{f(z_\ell)}{g'(z_\ell)} = c \sum_{\ell=1}^n (-1)^{n-\ell} \frac{f(z_\ell) \det V_\ell}{\det V} = \frac{c}{\det V} \sum_{\ell=1}^n (-1)^{n+\ell} f(z_\ell) \det V_\ell.$$

Then we see that $\sum_{\ell=1}^n (-1)^{n+\ell} f(z_\ell) \det V_\ell$ is the Laplace expansion along column n of the matrix

$$V_* = \begin{pmatrix} 1 & z_1 & z_1^2 & \cdots & z_1^{n-2} & f(z_1) \\ 1 & z_2 & z_2^2 & \cdots & z_2^{n-2} & f(z_2) \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & z_n & z_n^2 & \cdots & z_n^{n-2} & f(z_n) \end{pmatrix}.$$

Since $\deg f(x) \leq n-2$, the polynomial $f(x)$ can be written as $f(x) = a_{n-2}x^{n-2} + \cdots + a_1x^1 + a_0$ for some $a_{n-2}, \dots, a_0 \in \mathbb{C}$. In particular, $f(z_\ell) = a_{n-2}z_\ell^{n-2} + \cdots + a_1z_\ell^1 + a_0$ and thus $\det V_* = 0$. This completes the proof. $\square$

Remark that the solution (especially on determinant problems) requires some intuition and a bit of luck. As above, if we start by expanding the polynomial $g(x)$, then we terbabas and miss the Vandermonde matrix. Honestly I don't have good advice on how can we improve and reach the solution by "intuition". We can just practice to get some experiences, perhaps the techniques we have learned can be adapted to solve new problems.

Lastly, some other results for determinants are presented in case they can be used.

Proposition 1.4. 1. (Cramer's rule) Let $Ax = y$ be a linear system, where $A \in$

$M_n(\mathbb{F})$, $x, y \in M_{n,1}(\mathbb{F})$. If $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ and A is invertible, then

$$x_i = \frac{\det A(i; y)}{\det A}$$

where $A(i; y)$ is the matrix formed by replacing the i -th column of A by the vector y .

2. (Cauchy-Binet formula) Let $A \in M_{m,k}(\mathbb{F})$, $B \in M_{k,n}$ and $1 \leq p \leq \min\{m, k, n\}$. Let $I \subseteq \{1, \dots, m\}$ and $J \subseteq \{1, \dots, n\}$ be index sets with $|I| = |J| = p$. Then we have

$$\det((AB)[I, J]) = \sum_K \det A[I, K] \det B[K, J]$$

in which the sums are over all index sets $K \subseteq \{1, \dots, k\}$ with $|K| = p$.

1.6 Graphs and matrices

This topic is the one we may not realize the technique unless we are given a hint. It could very well come up in real competition! In this section, you should have some knowledge of graph theory since I couldn't write all the stuff. I encourage you to refer to the textbook *Graph Theory by J. A. Bondy and U. S. R. Murty*, which access is provided by UM Library!

Definition 1.15. Let $G = (V, E)$ be a graph (not necessarily undirected or simple) and let $V(G) = \{v_1, \dots, v_n\}$ be its (ordered) vertex set. The **adjacency matrix** of G is the $(0, 1)$ -matrix where

$$a_{ij} = \begin{cases} 1 & \text{if } v_i v_j \in E(G), \\ 0 & \text{if } v_i v_j \notin E(G). \end{cases}$$

Remark. If the notation is very confusing, we can actually just view the vertices and edges respectively as a set of integers $\{1, \dots, n\}$ and ordered pairs $\{(i, j) \in E(G)\}$. So

$$a_{ij} = \begin{cases} 1 & \text{if } (i, j) \in E(G), \\ 0 & \text{if } (i, j) \notin E(G). \end{cases}$$

Example. The adjacency matrix of an undirected simple graph is a symmetric $(0, 1)$ -matrix with trace zero. Note that the trace of the adjacency matrix denotes the number of loops in the graph. The converse is also true: Every symmetric $(0, 1)$ -matrix with trace zero defines an undirected simple graph.

Discussion 1.7 (Graph isomorphism and permutation matrix). Two graphs G and H are isomorphic if and only if there exists a bijection $\theta : V(G) \rightarrow V(H)$ such that such that $v_i v_j \in E(G)$ if and only if $\theta(v_i) \theta(v_j) \in E(H)$. Recall that a permutation matrix P has the property

$$P \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} x_{\psi(1)} \\ \vdots \\ x_{\psi(n)} \end{pmatrix}$$

for some bijection $\psi : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$. Let $A = (a_{ij})$ and $B = (b_{ij})$ be the adjacency matrices of G and H respectively (B is based on the ordering $\theta(v_1), \dots, \theta(v_n)$). Then G and H are isomorphic if and only if $(a_{ij}) = (b_{ij})$. Let $C = (c_{ij})$ be the adjacency matrix of H (ordering is not specified). We can just relabel the vertices to obtain B , i.e, there exists a bijection $\psi : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ such that $(b_{ij}) = (c_{\psi(i), \psi(j)})$. Since there exists a permutation matrix P such that

$$(c_{\psi(i), \psi(j)}) = P(c_{i, \psi(j)}) = P(c_{ij})P^T,$$

it follows that

$$A = PCP^T.$$

So the isomorphism of graphs is equivalent to similarity via a permutation matrix.

Example. A *topological sort* of a directed graph D is a linear ordering of its vertices such that, for every arc a of D , the tail of a precedes its head in the ordering. We can deduce (but not here) that a digraph admits a topological sort if and only if it is acyclic. In particular, the adjacency matrix of D with respect to the topological sort is a strictly upper triangular matrix. Hence any adjacency matrices of D can be permuted to a strictly upper triangular matrix.

Now we see an application of adjacency matrices in a combinatorial problem.

Discussion 1.8 (Counting the number of walks of given length). Let D be a directed graph with adjacency matrix A of order n . Then, for each positive integer k , the (i, j) -entry of A^k is the number of all walks in D of length k from vertex i to vertex j .

Sketch of proof. Use induction on k . Write $A^{k+1} = AA^k$ and use matrix multiplication to argue on the (i, j) -entry of AA^k . $\square$

Example. we can count the number of walks in the complete graph K_n . Let A be the adjacency matrix of K . Then we see that $A = J_n - I_n$, where J_n denotes the all-ones matrix. Note that $J_n^m = n^{m-1}J$ for all $m \geq 1$. Hence

$$\begin{aligned} A^k &= (J_n - I_n)^k = \sum_{i=0}^k \binom{k}{i} J_n^{k-i} (-I_n)^i = \sum_{i=0}^{k-1} \binom{k}{i} J_n^{k-i} (-I_n)^i + (-1)^k I_n \\ &= \sum_{i=0}^{k-1} \binom{k}{i} n^{k-i-1} (-1)^i J_n + (-1)^k I_n = \left(\sum_{i=0}^{k-1} \binom{k}{i} n^{k-i-1} (-1)^i \right) J_n + (-1)^k I_n \\ &= \left(\sum_{i=0}^k \binom{k}{i} n^{k-i-1} (-1)^i - n^{-1} (-1)^k \right) J_n + (-1)^k I_n \\ &= \left(n^{-1} \sum_{i=0}^k \binom{k}{i} n^{k-i} (-1)^i - n^{-1} (-1)^k \right) J_n + (-1)^k I_n. \end{aligned}$$

Finally, we get

$$A^k = \left(\frac{(n-1)^k - (-1)^k}{n} \right) J_n + (-1)^k I_n.$$

Example. Let G be an undirected simple graph with e edges and t triangles. Then we have

$$e = \frac{\text{tr } A^2}{2}, \quad t = \frac{\text{tr } A^3}{6}.$$

There is another matrix called the incidence matrix, defined for illustrating the relationship between sets (vertex set and edge set in graph theory).

Definition 1.16. Let D be a directed graph with $V(D) = \{v_1, \dots, v_n\}$ and $E(D) = \{e_1, \dots, e_m\}$. The **incidence matrix** M of D is given by

$$m_{ij} = \begin{cases} 1 & \text{if } v_i \text{ is the initial vertex of the edge } e_j, \\ -1 & \text{if } v_i \text{ is the terminal vertex of the edge } e_j, \\ 0 & \text{otherwise.} \end{cases}$$

The rows of M correspond to the vertices of D , and its columns correspond to the edges of D .

Based on the definition, we can observe that there is just one 1 and one -1 in each column, as it represents a directed edge from a vertex to a vertex. It follows that the sum of rows of M is zero vector, and thus the rows are linearly independent. This motivates the following result.

Theorem 1.3. Let c be the number of connected components of a directed graph D . If D has n vertices and M is the incidence matrix of D , then $\text{rank } M = n - c$.

Proof. Let $D_1, \dots, D_c$ be the connected components of D and n_i be the number of vertices in D_i for each i . Then, after relabelling of vertices and edges if necessary, we have

$$M = \begin{pmatrix} M_1 & 0 & \cdots & 0 \\ 0 & M_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & M_c \end{pmatrix}.$$

where M_i is the incidence matrix of D_i . It is sufficient to prove that $\text{rank } M_i = n_i - 1$, since

$$\text{rank } M = \sum_{i=1}^c \text{rank } M_i = \sum_{i=1}^c n_i - \sum_{i=1}^c 1 = n - c.$$

Fix $1 \leq i \leq c$. Let $M_i = \begin{pmatrix} r_1^{(i)} \\ \vdots \\ r_{n_i}^{(i)} \end{pmatrix}$, where $r_k^{(i)}$ is the k -th row of M_i for each $1 \leq k \leq n_i$.

Let $x = (x_1, \dots, x_{n_i}) \in M_{1, n_i}(\mathbb{F})$ be a nonzero vector such that $xM_i = 0$, in other words, $x^t \in \text{Ker } M_i^t$ (why x can be nonzero?). Then we have $x_1 r_1^{(i)} + \cdots + x_{n_i} r_{n_i}^{(i)} = 0$. Choose a row $r_k^{(i)}$ such that $x_k \neq 0$. This row (associated with a vertex k) has nonzero entries in the columns associated with the edges that are incident with k . In those columns, there is only one other nonzero entry with opposite sign in a row $r_\ell^{(i)}$ different from $r_k^{(i)}$. More precisely, if j -th entry of $r_k^{(i)}$ is nonzero, then comparing the j -th entry of the equality $x_1 r_1^{(i)} + \cdots + x_{n_i} r_{n_i}^{(i)} = 0$ gives $x_k = x_\ell$ for some $\ell \neq k$. So $x_k = x_\ell$ if k and ℓ are adjacent. Following the argument (now applied to vertex ℓ adjacent to k), we see that $x_k = x_\ell$ if there is an undirected (k, ℓ) -path. Since D_i is connected, it follows that $x_k^{(i)} = x_\ell^{(i)}$ for all $1 \leq \ell \leq n_i$. Hence $\text{Ker } M_i^t = \langle r_1^{(i)} + \cdots + r_{n_i}^{(i)} \rangle$. Consequently,

$$\text{rank } M_i = \text{rank } M_i^t = n_i - \text{null } M_i = n_i - 1$$

by rank-nullity theorem. □

We have seen the incidence matrix of a directed graph. Some may wonder how -1 changes when we define it for an undirected graph. In fact, we will see the edges as a *point set* and hence the incidence matrix is defined for a *block design*. Since it is just an introduction, I shall restrict it to the definition for subsets.

Definition 1.17. Let $X_1, \dots, X_m$ be subsets of an n -set $X = \{x_1, \dots, x_n\}$. The **incidence matrix** $A = (a_{ij})$ for the configuration of subsets is an $m \times n$ matrix with

$$a_{ij} = \begin{cases} 1 & \text{if } x_j \in X_i, \\ 0 & \text{otherwise.} \end{cases}$$

The 1's in the i -th row of A represent which elements are in the subset X_i .

Let χ_S be the **characteristic vector** of a subset S of X , i.e., $\chi_S = (a_1, \dots, a_n)$ where

$$a_k = \begin{cases} 1 & \text{if } x_k \in S, \\ 0 & \text{otherwise.} \end{cases}$$

Let T be another subset of X . The key observation is that $\chi_S \chi_T^t = |S \cap T|$. Based on the discussion, we can express the incidence matrix A as

$$A = \begin{pmatrix} \chi_{X_1} \\ \vdots \\ \chi_{X_m} \end{pmatrix}$$

and we get

$$AA^t = \begin{pmatrix} \chi_{X_1} \\ \vdots \\ \chi_{X_m} \end{pmatrix} (\chi_{X_1}^t \quad \cdots \quad \chi_{X_m}^t) = (\chi_{X_i} \chi_{X_j}^t) = (|X_i \cap X_j|).$$

It can be applied to the following problem.

Discussion 1.9 (Oddtown). Suppose there is a town with a population of n residents who love forming different clubs. To limit the number of possible clubs, the town council establishes the following rules:

1. Every club has an odd number of members.
2. Every two clubs must share an even number of members (possibly none).

Determine the maximum number of clubs.

Proof. Let $C_1, \dots, C_m$ be the clubs. We will show that $m \leq n$. Let A be the incidence matrix whose rows are indexed by the clubs and whose columns are indexed by the residents (ordered). By the working above, we have $AA^t = (|C_i \cap C_j|)$. If we work over $\mathbb{F}_2$ (indicating $|C_i \cap C_j|$ is even or odd), then, by the assumption, $AA^t = I_m$ (over $\mathbb{F}_2$). Hence

$$n \geq \min\{m, n\} \geq \text{rank } A \geq \text{rank}(AA^t) = \text{rank } I_m = m.$$

Clearly, the maximum can be achieved when each club contains only one member. $\square$

1.7 Problems

Problem 1.1 (MSE). Let V be a n -dimensional vector space over field $\mathbb{F}$. Let W be a subspace of V with dimension $p < n$. Show that W is the intersection of all $(n-1)$ -dimensional subspaces of V which contains W .

Problem 1.2 (No. 12, 2014 LA Example Sheet 1). Recall that $\mathbb{F}^n$ has standard basis $e_1, \dots, e_n$. Let U be a subspace of $\mathbb{F}^n$. Show that there is a subset I of $\{1, 2, \dots, n\}$ for which the subspace $W = \langle \{e_i : i \in I\} \rangle$ is a complementary subspace to U in $\mathbb{F}^n$.

Problem 1.3 (General version of Problem 1.2). Given two subsets $X = \{x_1, \dots, x_m\}$ and $Y = \{y_1, \dots, y_n\}$ of a vector space V over a field $\mathbb{F}$. Suppose that X is a linearly independent set and each vector in x can be written as a linear combination of Y . Show that there exists a permutation $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ such that $\langle x_1, \dots, x_k, y_{\sigma(k+1)}, y_{\sigma(k+2)}, \dots, y_{\sigma(n)} \rangle = \langle y_1, \dots, y_n \rangle$ for each $k = 1, \dots, m$. *Hint:* Read Steinitz exchange lemma.

Problem 1.4 (No. 13, 2014 LA Example Sheet 1). Suppose X and Y are linearly independent subsets of a vector space V ; no member of X is expressible as a linear combination of members of Y , and no member of Y is expressible as a linear combination of members of X . Is the set $X \cup Y$ necessarily linearly independent? Give a proof or counterexample.

Problem 1.5 (Ex. 24, Chap. 1, Advanced Linear Algebra). Let V be an n -dimensional vector space over $\mathbb{R}$.

- (i) If $n \geq 2$, show that there are infinitely many k -dimensional subspaces of V for each $k = 1, \dots, n-1$.
- (ii) Show that V is not the union of a finite number of proper subspaces.
- (iii) Suppose that $S_1, \dots, S_k$ are subspaces of V with $\dim(S_i) \leq m < n$. Prove that there is a subspace T of V of dimension $n-m$ for which $T \cap S_i = \{0\}$ for all i .

Problem 1.6 (MSE). Let $U_1, \dots, U_m$ be proper subspaces of an n -dimensional vector space V over $\mathbb{R}$.

- 1. Show that there is a basis $\{v_1, \dots, v_n\}$ of V outside the set $U_1 \cap \dots \cap U_m$.
- 2. Show that there is a basis $\{v_1, \dots, v_n\}$ of V outside the set $U_1 \cup \dots \cup U_m$.

Problem 1.7 (SMMC 2020 B1). Let $\mathcal{M}$ be the set of 5×5 real matrices of rank 3. Given a matrix A in $\mathcal{M}$, the set of columns of A has $2^5 - 1 = 31$ nonempty subsets. Let k_A be the number of these subsets that are linearly independent. Determine the maximum and minimum values of k_A , as A varies over $\mathcal{M}$. The rank of a matrix is the dimension of the span of its columns.

Problem 1.8 (Halmos's Linear Algebra problem book). A subset S of $\mathbb{R}^n$ with at least n vectors is called *relatively independent* if every n vectors in S are linearly independent. Show that there is no upper bound on the number of vectors in a relatively independent subset.

Problem 1.9 (SMMC 2023 B3). Let A , B and C be three n -dimensional subspaces of $\mathbb{R}^{2n}$ with the property that $A \cap B = B \cap C = C \cap A = \{0\}$. Prove that there exist bases $\{a_1, \dots, a_n\}$ of A , $\{b_1, \dots, b_n\}$ of B , and $\{c_1, \dots, c_n\}$ of C with the property that for each $i \in \{1, \dots, n\}$, the vectors a_i , b_i and c_i are linearly dependent.

Problem 1.10 (Putnam 1946 A2). If $a(x)$, $b(x)$, $c(x)$ and $d(x)$ are real polynomials in x , show that

$$\int_1^x a(t)c(t) dt \cdot \int_1^x b(t)d(t) dt - \int_1^x a(t)d(t) dt \cdot \int_1^x b(t)c(t) dt$$

is divisible by $(x-1)^4$.

Problem 1.11. Let V be a finite-dimensional vector space. Does there exist linear transformations $f, g \in \mathcal{L}(V)$ such that $f \circ g - g \circ f = \iota$? Here $\iota : V \rightarrow V$ denotes the identity map.

Problem 1.12. Let U , V and W be finite-dimensional vector spaces. Let $\tau \in \mathcal{L}(U, V)$ and $\phi \in \mathcal{L}(V, W)$. Suppose $M = \{\tau(x) : x \in U, (\phi \circ \tau)(x) = 0\}$. Show that $\dim M = \text{rank } \tau - \text{rank}(\phi \circ \tau)$.

Problem 1.13 (No. 6(i), 2024 LA Example Sheet 1). Let $\alpha : V \rightarrow V$ be a linear operator of a finite-dimensional vector space V . Show that

$$V \geq \operatorname{Im} \alpha \geq \operatorname{Im} \alpha^2 \geq \cdots.$$

If $r_k = \operatorname{rank} \alpha^k$, deduce that $r_k \geq r_{k+1}$ and $r_k - r_{k+1} \geq r_{k+1} - r_{k+2}$. Conclude that if, for some $k \geq 0$, we have $r_k = r_{k+1}$, then $r_k = r_{k+\ell}$ for all $\ell \geq 0$. *Remark:* This is another approach for Discussion 1.5.

Problem 1.14 (S.T. Yau College Math Contest Team Algebra 2018 P6). Let $\mathbb{F}$ be an algebraically closed field and let $m(x) \in \mathbb{F}[x]$ be the minimal polynomial of a matrix $A \in M_n(\mathbb{F})$, i.e., the unique monic polynomial of smallest degree such that $m(A) = 0$. Given a subspace $W_A = \{AX - XA : X \in M_n(\mathbb{F})\}$ of $M_n(\mathbb{F})$, prove that

$$\dim W_A \leq n^2 - \deg(m(x))$$

and the equality holds if and only if $\deg m(x) = n$.

Problem 1.15 (Somewhere in Functional Analysis textbook). Let V be a vector space over a field $\mathbb{F}$. Suppose that $f, f_1, \dots, f_n : V \rightarrow \mathbb{F}$ are linear functionals on V satisfying $\bigcap_{i=1}^n \operatorname{Ker} f_i \subseteq \operatorname{Ker} f$. Show that f is a linear combination of $f_1, \dots, f_n$.

Problem 1.16 (Ex. 23, Chap. 1, Advanced Linear Algebra). Let $\mathcal{B} = \{b_1, \dots, b_n\}$ and $\mathcal{C} = \{c_1, \dots, c_n\}$ be bases for a vector space V . Let $1 \leq m \leq n-1$. Show that there is a permutation σ of $\{1, \dots, n\}$ such that

$$b_1, \dots, b_m, c_{\sigma(m+1)}, \dots, c_{\sigma(n)}$$

and

$$c_{\sigma(1)}, \dots, c_{\sigma(m)}, b_{m+1}, \dots, b_n$$

are both bases for V . *Hint:* Use general Laplace expansion theorem.

Problem 1.17 (Desnanot-Jacobi identity). Let $A(i_1, \dots, i_s | j_1, \dots, j_t)$ denote the $(n-s) \times (n-t)$ submatrix of a matrix A obtained by deleting the rows indexed by $i_1, \dots, i_s$ and the columns indexed by $j_1, \dots, j_t$. Let M be an $n \times n$ matrix over a field such that $\det M(1, n | 1, n) \neq 0$. Show that

$$\det M = \frac{\det M(1|1) \det M(n|n) - \det M(1|n) \det M(n|1)}{\det M(1, n | 1, n)}.$$

Problem 1.18 (SMMC 2021 A3). Let $\mathcal{M}$ be the set of all 2021×2021 matrices with at most two entries in each row equal to 1 and all other entries equal to 0.

Determine the size of the set $\{\det A : A \in \mathcal{M}\}$.

Here $\det A$ denotes the determinant of the matrix A .

Problem 1.19 (Problem 3G, *A Course in Combinatorics*). Let m be given. Show that if n is large enough, every $n \times n$ $(0, 1)$ -matrix has a principal submatrix of size m , in which all the elements below the diagonal are the same, and all the elements above the diagonal are the same. *Hint:* Try edge-coloring and use a very famous theorem about the existence of certain graphs.

Remark: Here is the original statement for a problem in China TST 2005. "We call a matrix *binary matrix* if all its entries equal to 0 or 1. A binary matrix is *Good* if it simultaneously satisfies the following two conditions:

- (i) All the entries above the main diagonal (from left to right), not including the main diagonal, are equal.
- (ii) All the entries below the main diagonal (from left to right), not including the main diagonal, are equal.

Given positive integer m , prove that there exists a positive integer M , such that for any positive integer $n > M$ and a given $n \times n$ binary matrix A_n , we can select integers $1 \leq i_1 < i_2 < \cdots < i_{n-m} \leq n$ and delete the i_1 -th, i_2 -th, $\dots$, i_{n-m} -th rows and i_1 -th, i_2 -th, $\dots$, i_{n-m} -th columns of A_n , then the resulting binary matrix B_m is *Good*." It is the same problem, just described in a different way! So don't worry if the problem seems so lengthy, you may simplify the statement.

Problem 1.20. Given an $n \times n$ $(0, 1)$ -matrix with distinct rows, is it possible to delete a column so that the resulting matrix has distinct rows?

Problem 1.21 (4.3.7, Graph Theory by Bondy & Murty). Let G be a connected graph, and let M be its incidence matrix.

- a) Show that the columns of M corresponding to a subset S of the set of edges E of G are linearly independent over $\mathbb{F}_2$ if and only if $G[S]$ is acyclic.

Here $G[S]$ denotes the subgraph of G whose edge set is S and whose vertex set consists of all endvertices of edges of S .

- b) Deduce that there is a one-to-one correspondence between the bases of the column space of M over $\mathbb{F}_2$ and the spanning trees of G .

Problem 1.22. Suppose that n people live in Reverse Oddtown. Every club contains an even number of persons, and any two clubs share an odd number of persons. Show that no more than n clubs can be formed.

Problem 1.23 (IMC 2001 P4). Let $A = (a_{k,\ell})_{k,\ell=1,\dots,n}$ be an $n \times n$ complex matrix such that for each $m \in \{1, \dots, n\}$ and $1 \leq j_1 < \cdots < j_m \leq n$ the determinant of the matrix $(a_{j_k, j_\ell})_{k,\ell=1,\dots,m}$ is zero. Prove that $A^n = 0$ and that there exists a permutation $\sigma \in S_n$ such that the matrix

$$(a_{\sigma(k), \sigma(\ell)})_{k,\ell=1,\dots,n}$$

has all of its nonzero elements above the diagonal.

Problem 1.24 (Putnam 2021 B5). Say that an n -by- n matrix $A = (a_{ij})_{1 \leq i, j \leq n}$ with integer entries is *very odd* if, for every nonempty subset S of $\{1, 2, \dots, n\}$, the $|S|$ -by- $|S|$ submatrix $(a_{ij})_{i, j \in S}$ has odd determinant. Prove that if A is very odd, then A^k is very odd for every $k \geq 1$.